

Day 3: Windows + Linux + Command Line Foundations

Notes written by Kevin Beideman. Recommended time of study for these concepts to be understandable is approx 2hr 30min for beginners.

Why does Linux matter? [Linux in 100 Seconds](#)

Terminal Basics, Files and Directories, Windows vs Linux, Commands

~ is the home directory.

pwd (print working direction) shows the full location.

cd .. takes you to the root of the file system.

The ls command will list the files. It shows subdirectories and files within your current location.\

The touch command allows you to create a file. For example, touch hello.txt will create a new text file with that name.

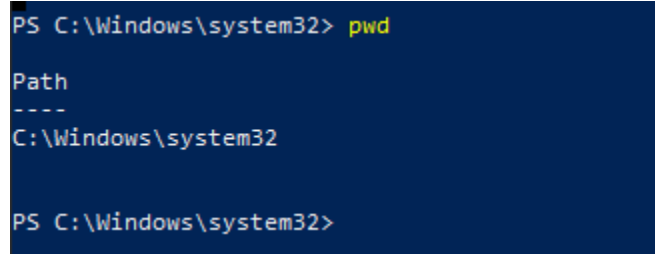
A tool like vi lets you edit that text file. You say vi hello.txt to enter and edit the file.

After this: say “cat hello.txt”. This will output the file’s contents to the terminal.

Saying grep -i “mom” is useful if the file is really big. The grep command lets you find text in a file.

Instructions: Command Line Training

Open Powershell or VS Code terminal. Type pwd. This means “Print Working Directory.” It shows where you currently are in the filesystem.



```
PS C:\Windows\system32> pwd

Path
----
C:\Windows\system32

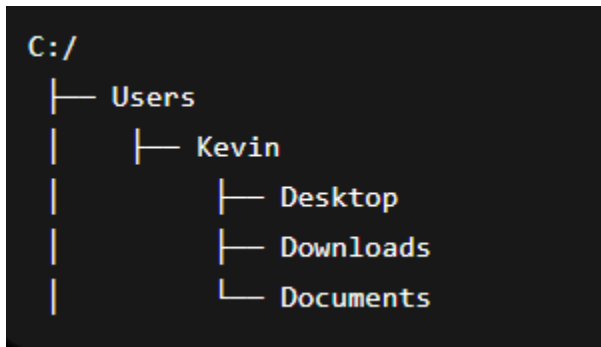
PS C:\Windows\system32>
```

Next, run ls. This command will list files and folders in the current directory. If you add on **ls -Force**, hidden items will be shown too.

The **cd** command lets you move between Directories. Try **cd Desktop**. If you type the variation, **cd ..**, you move one UP. Saying **cd ~**, returns to the home directory.

Inside Desktop, type **mkdir cyber-lab**. This will create a folder named “cyber-lab.” Move into the folder saying **cd cyber-lab**. To create a new item: **New-Item notes.txt**. Type **ls** now within the folder.

Understanding Filesystems



Everything is hierarchical. [Linux File System/Structure Explained!](#)

Task: Navigate manually to Desktop, Downloads, and Documents using only terminal commands.

Bonus: Run tree or tree /f

Next, try running **Get-Help cd** and **Get-Help ls**. This is how you can learn commands quickly, without memorization.